Manual

Interface Material and Batch Data

EIS-MCL 8.1

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Contents

1 Interface Material, Batch and Lot Data

Overview

Use options

The interface for material and batch data provides interface structures and functions, which allow for inventory information (or its extracts) to be exchanged between the higher level ERP/PPS level as the inventory-managing system and the MES level.

As material staging data is transferred from ERP/PPS to MES, it is possible to provide MES with the charges and batches available for production and to thus create the data reasons for plausibility checks upon the entry of material data.

In exchange, MES will provide information on produced materials (lots and/or batches) and on material consumptions for the ERP level.

Implementation notes

Use the interface for material, batch and lot data if you

- need information on provided lots/ batches as entry reasons in the context of tracking & tracing processes;
- wish to use information on produced materials as data base for materials management;
- wish to use information on material consumptions as data base for materials management;
- need the origin tree about inbound materials used for the production of materials for tracking purposes.

Integration

In the course of the data provision and data confirmation the interface for material, batch and lot data integrates into:

- Tracking & tracing where the provided data is accessed for plausibility checks;
- material and production logistics where the data provided by the interface will be further processed.

Scope of functions

- Transfer of material stagings:
  - Adoption of material stagings from the superior system to HYDRA
- Transfer of goods receipts:

    - Transfer of goods receipts for created lots and/or batches from HYDRA to the inventory-managing system according to different movement types
    - Confirmation of the origin tree at the goods receipt of a lot and/or batch
- Transfer of goods issues
    - Transfer of goods issues for consumed materials from HYDRA to the inventory-managing system with reference to the production order and different movement types
- Configuration of the interface

Configuration of the inventory-relevant material types in HYDRA to control the interface

2 Setup of Data Record Structure

The data are transferred in the following structure. Within this structure the value of the SEGNAM field precisely defines the set-up of the user data structure in the SDATA field.

Field name	Type	Length	Designation	Data field and meaning
SEGNAM*	Char	30	Segment name	This field is occupied by the writing system with the respective segment name. This precisely defines the set-up of the data record (SDATA field). Example: HY72_AU_HD_001
MANDT*	Char	3	Client	Reserved; fixed: '000'
DOCNUM*	Char	16	IDOC number	Serial number for the IDOCs Reserved: fixed '0000000000000000'
SEGNUM*	Char	6	Segment number	Reserved: fixed '000000'
PSEGNUM	Char	6	Parent segment number	Reserved; fixed: '000000'
HLEVEL	Char	2	Hierarchy level	Reserved; fixed: '00'
SDATA	Char	1000	User data	This field contains the user data. The structure of this field is defined by the SEGNAM field.

3 Data type definitions

Type	Description
CHAR x	Information is left-aligned for the data type CHAR. Places that are not required are filled with blanks (U+0020). If the field is not used, it must be completely prepopulated with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. The data type NUMC only supports digits (ASCII characters 30 Hex to 39 Hex). These digits are right-aligned and unnecessary places are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030). Example: "00000002"
DEC x.y **QUAN x.y**	Numeric field of the length x and y decimal places. A data field in HYDRA format is **preceded** by a sign ("+" or "-") and includes a decimal point. Places that are not required are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030) including algebraic sign and decimal separator. e.g. DEC 13,3: • `-1234567890,123` → `-1234567890.123` • `234567890,3` → `+0234567890.300` Note: The field length is indicated WITHOUT algebraic sign and WITHOUT decimal point in the tabular description of the structure. This means, for example, that a field QUAN 13.3 is converted to an external length of CHAR15.
DATE	Format YYYYMMDD. If the field is not used, it must remain empty (filled with blanks (U+0020).
TIME	Format HHMMSS. If the field is not used, it must be set to “000000” (zeros with (U+0030)).

HYDRA does not support special characters for all alphanumeric fields. This includes, among others: "\“ (backslash), "|“ (pipe), „ “ “ (double quotes), and " ’ “ (single quotes). You cannot enter these characters using shop floor terminals and the MOC does not support them.

The character " **;** " (semicolon) is used as a separator in the system. You must not use this character in key fields (e.g. order/operation number, MES batch number, personnel number).

The character " % " (percent) is used as a placeholder for database communication. You should not use this character to prevent the result from being falsified.

4 Material Supply ERP --> MES

Overview

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Certain material movements are transferred from the ERP system to the MES. IDocs are created cyclically (e.g. every 5 minutes) and transferred to HYDRA. IDocs are called ZMBEW03.

The MES material type must also be stored in the ERP system and transferred together with the provided data.

If batch movements are transferred including user-specific information, this information must also be transferred to HYDRA.

This leads to the following IDoc specification:

Message type/ file name:	**ZMBEW**	
File extension (for file transfer)	**DAT**	
IDoc type (for tRFC communication):	**ZMBEW03**	
Segments:	**Z2MBEW001X000** (goods movements)	1 - n
	Z2MBEW002X000 (goods movements continued)[1]	0 - 1
	Z2CNRATT_C000X000 (alphanumeric batch attributes part 1)	0 - 1
	Z2CNRATT_C001X000 (alphanumeric batch attributes part 2)	0 - 1
	Z2CNRATT_N000X000 (numeric attributes) – (OBSOLETE)[2]	0 – 1
	Z2CNRATT_N001X000 (numeric attributes)[3]	0 - 1
	Z2CNR_USRFLD000X000 (user fields)[4]	0 - 1

Segments in SAP have to be created according to the pattern Z1<segment name> in order to generate the above-mentioned segment names in SAP. The version control function of SAP outbound processing generates segment names in the form Z2<segment name><version>.

Example: segment name **Z1**MBEW001X is converted to **Z2**MBEW001X**000**

This documentation uses the below-mentioned column headings with the meaning described here:

Column	Description
Field	Field name
V (usage)	S Key field clearly identifying the data record. (Further key fields might be required). The field must be completed. M Mandatory field which must be filled with a valid value. K Field may stay empty (optional field).
T(ype)	Data type according to definition
L(ength)	Field length For fields of data type DEC: Overall number of digits without decimal separator and algebraic sign
D(ecimal places)	For fields of data type DEC: Number of decimal places; otherwise: not relevant

[1] The segment is available from MLE option ZMBEW_012 on.

[2] We recommend to no longer use this segment for new implementations. Segment Z2CNRATT_N001X000 should be used instead. Segment Z2CNRATT_N000X000 is still available (backwards compatible) but will no longer be maintained. Both segments have different field lengths for their decimal fields.

[3] The segment is available from MLE option ZMBEW_014 on.

[4] The segment is available from MLE option ZMBEW_010 on.

Column	Description
Description	Description and/or comment of the field

Goods movements (Z2MBEW001X000)

Material supplies: Goods receipt, reposting to a different material buffer. If a batch is already known in HYDRA, the batch inventory will be updated.

The batch inventory can only be updated if the batch is currently not in use (not running), i.e. batch status <> "L"

Field	V	T	L	D	Description
WERK	K	CHAR	4		Company/ plant
MATNR	M	CHAR	40		Material number
MATTYP	M	CHAR	10		Material type
MATTXT	K	CHAR	40		Material description
MATPUF	M	CHAR	12		Target storage location: Material buffer including the material; in case the material is transferred, the new material buffer is indicated here.
LAGORT	K	CHAR	20		PPS storage location
LAGPZ	K	CHAR	20		PPS storage bin
MNR	K	CHAR	10		Workplace which incoming goods are explicitly provided for (reserved, should be left empty)
CHARGE	K	CHAR	10		ERP batch number For further information on the meaning of the fields CHARGE and CHARGE_LONG please see the description below.
HY_LOSNR	S	CHAR	20		HYDRA batch number (to find a batch created from HYDRA)
MENGE	K	QUAN	13	3	Batch quantity
MENGE_EINH	K	CHAR	3		Batch quantity unit
LST01	K	QUAN	13	3	Activity 1
LST01_EINH	K	CHAR	3		Unit of activity 1
LST02	K	QUAN	13	3	Activity 2
LST02_EINH	K	CHAR	3		Unit of activity 2

Field	V	T	L	D	Description
LST03	K	QUAN	13	3	Activity 3
LST03_EINH	K	CHAR	3		Unit of activity 3
LST04	K	QUAN	13	3	Activity 4
LST04_EINH	K	CHAR	3		Unit of activity 4
LST05	K	QUAN	13	3	Activity 5
LST05_EINH	K	CHAR	3		Unit of activity 5
LST06	K	QUAN	13	3	Activity 6
LST06_EINH	K	CHAR	3		Unit of activity 6
LOSHINWEIS	K	CHAR	20		Entered info on batch
KLASSE	K	CHAR	1		Batch class "G" Yield "A" Scrap/ waste
STATUS	K	CHAR	1		Batch status to be set "F" Free/available "S" Blocked "D" Deleted
MATST	K	CHAR	1		Material status
QST	K	CHAR	1		Q status
QSTMANU	K	CHAR	1		Manual Q status
TST	K	CHAR	1		Transport status F - "normal" HYDRA batch B - Ready for booking out L - Cleared from stock O - Booked out for third-parties I - Sent to third-parties T - Transport done
GRUND	K	NUM	4		Batch reason (e.g. blocking reason)
GRUNDTYP	K	CHAR	1		Reason type (e.g. L - batch reason) Please note: Currently not used.
VDAT	K	DATE	8		Availability date (MPL-PUE) If MPL-PUE is not used, the value should be set to the current point in time.
VVZEI	K	TIME	6		Availability time (MPL-PUE) If MPL-PUE is not used, the value should be set to the current point in time.

Field	V	T	L	D	Description
WDAT	K	DATE	8		Warning date (MPL-PUE) If MPL-PUE is not used, the value should be set to 31.12.9999.
WZEI	K	TIME	6		Warning time (MPL-PUE) If MPL-PUE is not used, the value should be set to 11.59.59 p.m.
VFDAT	K	DATE	8		Expiry date (MPL-PUE) If MPL-PUE is not used, the value should be set to 31.12.9999.
VFZEI	K	TIME	6		Expiry time (MPL-PUE) If MPL-PUE is not used, the value should be set to 11.59.59 p.m.
RFAGVFA	K	DEC	13	3	Mass per unit area, unit G/ M2 otherwise: 0 Please note: Quantities will not automatically be converted
RFBREITE	K	DEC	13	3	Width of the roll in the unit MM otherwise: 0
CNR:RFSTKF	K	DEC	13	3	Area per piece in the unit MM2/ PCE otherwise: 0
RESART	K	CHAR	2		Reservation type: "AK" for order "AG" for subsequent operation "CC" for planning (comment)
RESVAL	K	CHAR	40		Reservation for order/ OP (only if RES:ART = "AK" or "AG")
RESBEM	K	CHAR	100		Comment about reservation (only if RES:ART = "CC")
ALT1	K	CHAR	20		Alternative batch number 1
ALT2	K	CHAR	20		Alternative batch number 2
ALT3	K	CHAR	20		Alternative batch number 3
ALT4	K	CHAR	20		Alternative batch number 4
ALT5	K	CHAR	40		Alternative batch number 5
EXTCNR	K	CHAR	20		External batch number
TPE	K	CHAR	10		Transport unit
TECHINFO	K	CHAR	20		Technical info
MCNR	K	CHAR	20		Merged batch number
SLOS	K	CHAR	1		Y - Merged batch Only relevant for merged batch processing
SLOSTYP	K	CHAR	1		Merged batch type 'J' – equal type Only relevant for merged batch processing
ANZSLOS	K	NUM	8		Number of individual batches included in merged batch Only relevant for merged batch processing
TRANZ	K	NUM	8		Number of batches Currently not used

Field	V	T	L	D	Description
TRANZSUM	K	NUM	8		Currently not used
LOSINDEX	K	NUM	8		Batch index Currently not used
TRPOS	K	NUM	8		Currently not used
SCHNEIDNR	K	NUM	8		Currently not used
ATTR1	K	NUM	8		Direct batch attribute 1
ATTR2	K	NUM	8		Direct batch attribute 2
ATTR3	K	NUM	8		Direct batch attribute 3
ATTR4	K	DEC	13	3	Direct batch attribute 4
ATTR5	K	DEC	13	3	Direct batch attribute 5
ATTR6	K	DEC	13	3	Direct batch attribute 6
ATTR7	K	CHAR	4		Direct batch attribute 7
ATTR8	K	CHAR	10		Direct batch attribute 8
ATTR9	K	CHAR	10		Direct batch attribute 9
ATTR10	K	CHAR	20		Direct batch attribute 10
HSDAT	K	DATE	8		Manufacturing date (available starting with MLE option ZMBEW_008)
HSZEI	K	TIME	6		Manufacturing time (available starting with MLE option ZMBEW_008)
CSTWDAT	K	DATE	8		Last change of status - date (available starting with MLE option ZMBEW_008)
CSTWZEI	K	TIME	6		Last change of status - time (available starting with MLE option ZMBEW_008)
CHARGE_LONG	K	CHAR	20		ERP batch number (long) For further information on the meaning of the fields CHARGE and CHARGE_LONG please see the description below. (from MLE option ZMBEW_012 on)

Please note that depending on the material type, individual OP-based user fields (transferred together with the operation from the PPS system) can be transferred to batch-related user fields (so called batch attributes) during the generation of batches in HYDRA. These are then transferred to the PPS system as ERP batch together with the batch at the goods receipt and will be returned when such an ERP batch is provided the next time.

Information on the fields CHARGE and CHARGE_LONG

The following logic/condition applies for providing the SAP batch:

Checking if the CHARGE (CHAR10) field is assigned a value:

- Yes:

  This value is adopted.

- No:

  Checking if the CHARGE_LONG (CHAR20) field is completed:

  - Yes:

    This value is adopted as PPS batch number.

  - No:

    This value ("nothing") is adopted as PPS batch number.

Goods movements continued (Z2MBEW002X000)[5]

The batch inventory can only be updated if the batch is currently not in use (not running), i.e. batch status <> "L"

Field	V	T	L	D	Description
HY_LOSNR	M	CHAR	20		HYDRA batch number (to find a batch created from HYDRA)
HULEVEL	K	NUMC	8		HU level
DLL	K	CHAR	20		External batch number (throughput batch number)
BESTART	K	CHAR	1		Stock type
MSL_VFDATE	K	DATE	8		MSL Expiry date (from MLE option ZMBEW_015 on)
MSL_VFTIME	K	TIME	6		MSL Expiry time (from MLE option ZMBEW_015 on)
MSL_PERIOD	K	NUMC	13		MSL term (from MLE option ZMBEW_015 on)

[5] The segment is available from the MLE option ZMBEW_012 on.

Alphanumeric batch attributes part 1 (Z2CNRATT_C000X000)

The following segment supports the (first 20) alphanumeric batch attributes for the transfer.

Field	V	T	L	D	Description
CNR	S	CHAR	20		HYDRA batch number
ATTR:101	K	CHAR	40		Alphanumeric attribute
ATTR:102	K	CHAR	40		…
ATTR:103	K	CHAR	40		…
ATTR:104	K	CHAR	40		…
ATTR:105	K	CHAR	40		…
ATTR:106	K	CHAR	40		…
ATTR:107	K	CHAR	40		…
ATTR:108	K	CHAR	40		…
ATTR:109	K	CHAR	40		…
ATTR:110	K	CHAR	40		…
ATTR:111	K	CHAR	40		...
ATTR:112	K	CHAR	40		...
ATTR:113	K	CHAR	40		...
ATTR:114	K	CHAR	40		...
ATTR:115	K	CHAR	40		...
ATTR:116	K	CHAR	40		...
ATTR:117	K	CHAR	40		...
ATTR:118	K	CHAR	40		...
ATTR:119	K	CHAR	40		...
ATTR:120	K	CHAR	40		…

Alphanumeric batch attributes part 2 (Z2CNRATT_C001X000)

The following segment supports the (last 20) alphanumeric batch attributes for the transfer.

Field	V	T	L	D	Description
CNR	S	CHAR	20		HYDRA batch number
ATTR:121	K	CHAR	40		Alphanumeric attribute
ATTR:122	K	CHAR	40		…
ATTR:123	K	CHAR	40		…
ATTR:124	K	CHAR	40		…
ATTR:125	K	CHAR	40		…
ATTR:126	K	CHAR	40		…
ATTR:127	K	CHAR	40		…
ATTR:128	K	CHAR	40		…
ATTR:129	K	CHAR	40		…
ATTR:130	K	CHAR	40		…
ATTR:131	K	CHAR	40		...
ATTR:132	K	CHAR	40		...
ATTR:133	K	CHAR	40		...
ATTR:134	K	CHAR	40		...
ATTR:135	K	CHAR	40		...
ATTR:136	K	CHAR	40		...
ATTR:137	K	CHAR	40		...
ATTR:138	K	CHAR	40		...
ATTR:139	K	CHAR	40		...
ATTR:140	K	CHAR	40		…

Numeric batch attributes (Z2CNRATT_N000X000) – (OBSOLETE)

We recommend to no longer use this segment for new implementations. Segment Z2CNRATT_N001X000 should be used instead. Segment Z2CNRATT_N000X000 is still available (backwards compatible) but will no longer be maintained. Both segments have different field lengths for their decimal fields.

The following segment supports numeric batch attributes for the transfer.

Field	L	T	L	D	Description
CNR	S	CHAR	20		HYDRA batch number
ATTR:201	K	NUMC	8		Numeric attribute, integer
ATTR:202	K	NUMC	8		...
ATTR:203	K	NUMC	8		...
ATTR:204	K	NUMC	8		...
ATTR:205	K	NUMC	8		...
ATTR:206	K	NUMC	8		...
ATTR:207	K	NUMC	8		...
ATTR:208	K	NUMC	8		...
ATTR:209	K	NUMC	8		...
ATTR:210	K	NUMC	8		...
ATTR:211	K	NUMC	8		...
ATTR:212	K	NUMC	8		...
ATTR:213	K	NUMC	8		...
ATTR:214	K	NUMC	8		...
ATTR:215	K	NUMC	8		...
ATTR:216	K	NUMC	8		...
ATTR:217	K	NUMC	8		...
ATTR:218	K	NUMC	8		...
ATTR:219	K	NUMC	8		...
ATTR:220	K	NUMC	8		...
ATTR:301	K	DEC	10	3	Numeric attribute, decimal
ATTR:302	K	DEC	10	3	...
ATTR:303	K	DEC	10	3	...
ATTR:304	K	DEC	10	3	...
ATTR:305	K	DEC	10	3	...
ATTR:306	K	DEC	10	3	...
ATTR:307	K	DEC	10	3	...
ATTR:308	K	DEC	10	3	...
ATTR:309	K	DEC	10	3	...
ATTR:310	K	DEC	10	3	...
ATTR:311	K	DEC	10	3	...
ATTR:312	K	DEC	10	3	...
ATTR:313	K	DEC	10	3	...
ATTR:314	K	DEC	10	3	...
ATTR:315	K	DEC	10	3	...
ATTR:316	K	DEC	10	3	...
ATTR:317	K	DEC	10	3	...
ATTR:318	K	DEC	10	3	...
ATTR:319	K	DEC	10	3	...
ATTR:320	K	DEC	10	3	...

Numeric batch attributes (Z2CNRATT_N001X000)

The following segment supports numeric batch attributes for the transfer.

Field	L	T	L	D	Description
CNR	S	CHAR	20		HYDRA batch number
ATTR:201	K	NUMC	8		Numeric attribute, integer
ATTR:202	K	NUMC	8		...
ATTR:203	K	NUMC	8		...
ATTR:204	K	NUMC	8		...
ATTR:205	K	NUMC	8		...
ATTR:206	K	NUMC	8		...
ATTR:207	K	NUMC	8		...
ATTR:208	K	NUMC	8		...
ATTR:209	K	NUMC	8		...
ATTR:210	K	NUMC	8		...
ATTR:211	K	NUMC	8		...
ATTR:212	K	NUMC	8		...
ATTR:213	K	NUMC	8		...
ATTR:214	K	NUMC	8		...
ATTR:215	K	NUMC	8		...
ATTR:216	K	NUMC	8		...
ATTR:217	K	NUMC	8		...
ATTR:218	K	NUMC	8		...
ATTR:219	K	NUMC	8		...
ATTR:220	K	NUMC	8		...
ATTR:301	K	DEC	13	3	Numeric attribute, decimal
ATTR:302	K	DEC	13	3	...
ATTR:303	K	DEC	13	3	...
ATTR:304	K	DEC	13	3	...
ATTR:305	K	DEC	13	3	...
ATTR:306	K	DEC	13	3	...
ATTR:307	K	DEC	13	3	...
ATTR:308	K	DEC	13	3	...
ATTR:309	K	DEC	13	3	...
ATTR:310	K	DEC	13	3	...
ATTR:311	K	DEC	13	3	...
ATTR:312	K	DEC	13	3	...
ATTR:313	K	DEC	13	3	...
ATTR:314	K	DEC	13	3	...
ATTR:315	K	DEC	13	3	...
ATTR:316	K	DEC	13	3	...
ATTR:317	K	DEC	13	3	...
ATTR:318	K	DEC	13	3	...
ATTR:319	K	DEC	13	3	...
ATTR:320	K	DEC	13	3	...

User fields (Z2CNR_USRFLD000X000)

Starting as of MLE option ZMBEW_010, you can also transfer user fields relating to the batch from the ERP system.

Field	V	T	L	D	Description
HY_LOSNR	S	CHAR	20		HYDRA batch number
FILLER	S	CHAR	20		Placeholder – internal usage
USRFLD	S	CHAR	8		User field key
FU:1	K	DATE	8		User field 1
FU:2	K	DATE	8		User field 2
FU:3	K	DATE	8		User field 3
FU:4	K	DATE	8		User field 4
FU:5	K	DATE	8		User field 5
FU:6	K	DATE	8		User field 6
FU:7	K	NUM	8		User field 7
FU:8	K	NUM	8		User field 8
FU:9	K	NUM	8		User field 9
FU:10	K	NUM	8		User field 10
FU:11	K	NUM	8		User field 11
FU:12	K	NUM	8		User field 12
FU:13	K	NUM	8		User field 13
FU:14	K	NUM	8		User field 14
FU:15	K	NUM	8		User field 15
FU:16	K	NUM	8		User field 16
FU:17	K	NUM	8		User field 17
FU:18	K	NUM	8		User field 18
FU:19	K	NUM	8		User field 19
FU:20	K	NUM	8		User field 20
FU:21	K	NUM	8		User field 21
FU:22	K	NUM	8		User field 22
FU:23	K	DEC	13	3	User field 23
FU:24	K	DEC	13	3	User field 24
FU:25	K	DEC	13	3	User field 25
FU:26	K	DEC	13	3	User field 26
FU:27	K	DEC	13	3	User field 27
FU:28	K	DEC	13	3	User field 28
FU:29	K	CHAR	1		User field 29
FU:30	K	CHAR	1		User field 30
FU:31	K	CHAR	1		User field 31
FU:32	K	CHAR	1		User field 32
FU:33	K	CHAR	1		User field 33
FU:34	K	CHAR	1		User field 34
FU:35	K	CHAR	1		User field 35
FU:36	K	CHAR	1		User field 36
FU:37	K	CHAR	1		User field 37
FU:38	K	CHAR	1		User field 38

Field	V	T	L	D	Description
FU:39	K	CHAR	1		User field 39
FU:40	K	CHAR	1		User field 40
FU:41	K	CHAR	1		User field 41
FU:42	K	CHAR	1		User field 42
FU:43	K	CHAR	1		User field 43
FU:44	K	CHAR	1		User field 44
FU:45	K	CHAR	10		User field 45
FU:46	K	CHAR	10		User field 46
FU:47	K	CHAR	10		User field 47
FU:48	K	CHAR	10		User field 48
FU:49	K	CHAR	10		User field 49
FU:50	K	CHAR	10		User field 50
FU:51	K	CHAR	20		User field 51
FU:52	K	CHAR	20		User field 52
FU:53	K	CHAR	20		User field 53
FU:54	K	CHAR	20		User field 54
FU:55	K	CHAR	20		User field 55
FU:56	K	CHAR	20		User field 56
FU:57	K	CHAR	20		User field 57
FU:58	K	CHAR	20		User field 58
FU:59	K	CHAR	20		User field 59
FU:60	K	CHAR	20		User field 60
FU:61	K	CHAR	20		User field 61
FU:62	K	CHAR	20		User field 62
FU:63	K	CHAR	20		User field 63
FU:64	K	CHAR	20		User field 64
FU:65	K	CHAR	40		User field 65
FU:66	K	CHAR	40		User field 66

5 Material consumptions MES --> ERP

Overview

The system transfers the material withdrawals recorded in the MES including the return transfers to the ERP system.

In the ERP system the withdrawal postings are deducted from the warehouse and added to the individual production order. If withdrawals for an ERP batch cannot be executed in the production warehouse specified by the MES (missing inventory), the posting will be performed for that warehouse where the ERP batch is found. Only if this also fails will the system issue an error message. The ERP system should also use this approach (searching for an ERP batch) for return transfers.

The MES provides this data to the ERP system at regular intervals. The IDoc is of the type **ZWAU02**. This leads to the following IDoc specification:

Message type	**ZWAU**
File name (for file transfers)	**Z2WAU000X000**
File extension (for file transfer)	Depending on the configuration (by default ".dat“)
IDOC type (with tRFC communication):	**ZWAU02**
Segments:	**Z2WAU000X000** (goods issues)

Respect the following conventions if SAP is connected:

Create SAP segment names according to the pattern Z1<segment name> in order to generate the above-mentioned segment names in SAP. Versioning in SAP outbound processing then creates the segment names in the form Z2<Segment name><Version>.

Example: the created segment name **Z1**WAU000X is converted to **Z2**WAU000X**000.**

This documentation uses the below-mentioned column headings with the meanings described here:

Column	Description	
Field	Name of the field	
V (usage)	S	Key field clearly identifying the data record. (Further key fields might be required). The field must be populated.
	M	Mandatory field which must be populated with a valid value.
	K	Field may stay empty (optional field).

Column	Description
T(ype)	Data type according to description.
L(ength)	Field length For fields of data type DEC: Total number of digits without decimal separator and algebraic sign.
D(ecimal places)	For fields of data type DEC: Number of decimal places; otherwise: not relevant
Description	Field description and/or comments on the field.

Movement types

In general, the following movement types are supported:

Movement type	Description/ usage
261	Goods issue for production order
262	Cancellation of goods issue for production order

Data structure (Z2WAU000X000)

Field	T	L	D	Description
WERK	CHAR	4		Company/ plant/ site; stored in MES (fixed)
BEWART	CHAR	3		Movement type (see table above)
MATPUF	CHAR	12		referring to batch: Material buffer according to batch inventory in MES anonymous batch → input buffer of the machine number (MNR)
LGORT	CHAR	20		ERP storage location Storage location stored to the material buffer.
ATK	CHAR	40		Material number
MENGE	QUAN	13	3	Batch quantity/ consumption The consumed quantity is always provided with a positive sign.
MENGE_EINH	CHAR	3		Unit for batch quantity/consumption
ANR	CHAR	40		Combined HYDRA production order number for which material has been withdrawn. The exact length that is uploaded/confirmed depends on how the lengths are configured for the order or operation in the HYDRA basic parameter settings. Used for ERP inbound processing if SAP is not used.
SAP_AUNR	CHAR	12		SAP order number Used for ERP inbound processing if SAP is in use.
SAP_AFOLG	CHAR	6		SAP sequence number Used for ERP inbound processing if SAP is in use.
SAP_VORNR	CHAR	4		SAP operation number Used for ERP inbound processing if SAP is in use.
SAP_UVGNR	CHAR	4		SAP sub-operation number Used for ERP inbound processing if SAP is in use.
KZEAUS	CHAR	1		J - indicates whether this is the last withdrawal for this component. The system sets this ID for automatically logged off input batches when logging off operations. The ID indicates that no further consumption occurs for an operation-related component.
HY_LOSNR	CHAR	20		MES batch number (corresponding to this ERP batch).

Field	T	L	D	Description
HY_DLLNR	CHAR	20		MES throughput batch number (corresponding to this ERP batch)
ARBPL	CHAR	8		Consuming MES machine
GRND	NUM	4		Reserved (e.g. blocking reason when batch is logged off)
LHW	CHAR	20		Information on batch if there is a reference to the input batch
PERSNO	NUM	8		Personnel number of the person logged in to the terminal
LST01	QUAN	13	3	Consumption activity 1
LST01_EINH	CHAR	3		Unit of activity 1
LST02	QUAN	13	3	Consumption activity 2
LST02_EINH	CHAR	3		Unit of activity 2
LST03	QUAN	13	3	Consumption activity 3
LST03_EINH	CHAR	3		Unit of activity 3
LST04	QUAN	13	3	Consumption activity 4
LST04_EINH	CHAR	3		Consumption unit of activity 4
LST05	QUAN	13	3	Consumption activity 5
LST05_EINH	CHAR	3		Unit of activity 5
LST06	QUAN	13	3	Consumption activity 6
LST06_EINH	CHAR	3		Unit of activity 6
LOSSTATUS	CHAR	1		Batch status when input batch is logged off
LOSKLASSE	CHAR	1		Batch class when input batch is logged off
CHARGENNUMMER	CHAR	10		ERP batch number
CHARGENNUMMER_LONG	CHAR	20		ERP batch number (long) Available from MPL 8.2 on - please also see the following information on field CHARGENNUMMER_LONG
MSL_VFDATE	DATE	8		MSL expiry date This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc“ is executed.
MSL_VFTIME	TIME	6		MSL expiry time This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc“ is executed.
MSL_PERIOD	NUMC	8		MSL term This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc“ is executed.

Information on the fields CHARGENNUMMER / CHARGENNUMMER_LONG

The field CHARGENNUMMER_LONG is only available as of MPL 8.2.

If MPL 8.2 is used, the fields CHARGENNUMMER and CHARGENNUMMER_LONG are populated as follows:

CHARGENNUMMER includes the ERP batch number with the characters 1-10.

CHARGENNUMMER_LONG includes the ERP batch number with the characters 1-20.

You should use the value of the CHARGENNUMMER_LONG field for new installations.

6 Goods Receipt MES --> ERP

Overview

	→	
	→	

The interface transfers all output batches generated from production orders in the MES for inventory-managed material maintained in the ERP system to the ERP system.

The interface transfers the tree of origin and the user-defined fields (of the producing operation) for each lot (= batch).

The MES provides the data at regular intervals to the ERP system. The IDoc is of the type ZWEI02. This leads to the following specification:

Message type:	**ZWEI**	
File name (for file transfers)	**Z2WEI000X000**	
File extension (for file transfers):	Depending on the configuration (by default ".dat")	
IDOC type (with tRFC communication):	**ZWEI02**	
Segments:	**Z2WEI000X000** (goods receipts)	1 – n
	Z2CNRATT_C000X000 (alphanumeric batch attributes part 1) 0 – 1	
	Z2CNRATT_C001X000 (alphanumeric batch attributes part 2) 0 – 1	
	Z2CNRATT_N000X000 (numeric batch attributes) - OBSOLETE[6]	0 – 1
	Z2CNRATT_N001X000 (numeric batch attributes)[7]	0 – 1
	Z2CNRBAUM000X000 (optional: tree of generation)	0 – n
	Z2TOLO000X000 (optional: sub-batches)	0 – n
	Z2CNR_USRFLD000X000 (optional: user fields)[8] 0 – 1	

6 You should no longer use this segment for new installations. Use the segment Z2CNRATT_N001X000 instead. The segment Z2CNRATT_N000X000 is still available (backwards compatible) but will no longer be maintained. Both segments have different field lengths for their decimal fields.

7 Please note the information on the activation in section Numeric attributes (Z2CNRATT_N001X000).

8 Please note the information on the activation in section User fields (Z2CNR_USRFLD000X000).

Note the following conventions if SAP is connected:

Create SAP segment names according to the pattern Z1<segment name> in order to generate the above-mentioned segment names in SAP. Versioning in SAP outbound processing then generates segment names in the form Z2<Segment name><Version>.

Example: the created segment name **Z1**WEI000X becomes **Z2**WEI000X**000**

Movement types

Goods receipts are transferred from the MES to the ERP system using the IDoc and message types described below. The movement type specifies the action (goods receipt, stock transfer, etc.) that must be triggered in the ERP system.

The following table describes which movement type is used to transfer goods receipt postings to the ERP system:

Movement type	Description/ usage
101	Standard goods receipt from production order: Goods receipt for batches where the option "Transfer to interface“ is set for the material type.
102	Cancellation of goods receipt
525	Goods receipt for batches that are blocked in the MES.
531	Goods receipt for waste batches
532	Cancellation of goods receipt for waste batches and batches

Goods receipt (Z2WEI000X000)

The data record described below transfers the goods receipt. The detailed information:
- user-specific fields
- tree of generation
- sub-batches for pallets
is transferred in sub-segments.

Field	T	L	D	Description
WERK	CHAR	4		Plant; stored in HYDRA (fixed)
BEWART	CHAR	3		Movement type: *see table above*
GRUND	NUM	4		e.g. blocking reason for BEWART 525

Field	T	L	D	Description
ZLO	CHAR	12		Target material buffer
LGORT	CHAR	20		ERP storage location Storage location stored to the target material buffer.
LGPZ	CHAR	20		ERP storage bin
CHARGE	CHAR	10		ERP batch number
MATNR	CHAR	40		Material number
MATTYP	CHAR	10		Material type in HYDRA
MENGE	QUAN	13	3	Quantity (primary quantity of the producing operation)
MENGE_EINH	CHAR	3		Quantity unit (primary quantity of the producing operation) Unit for the quantity in MENGE
ANR	CHAR	40		HYDRA order number = combined order/ operation number The exact length that is uploaded/confirmed depends on how the lengths are configured for the order or operation in the HYDRA basic parameter settings. Used for ERP inbound processing if SAP is not in use.
SAP_AUNR	12			SAP order number Used for ERP inbound processing if SAP is in use.
SAP_AFOLG	6			SAP sequence number Used for ERP inbound processing if SAP is in use.
SAP_VORNR	4			SAP operation number Used for ERP inbound processing if SAP is in use.
SAP_UVGNR	4			SAP sub-operation number Used for ERP inbound processing if SAP is in use.
KZSOBEST	CHAR	1		Not used, populated with " " (blanks).
KZ_ABZWEIG	CHAR	1		Indicates if the material derives from the planned deducted material: Comes from the OP data record and has the following meaning: "M“ Master OP of planned deducted material "K“ Sub-OP of planned deducted material " “ other
END_LIEF	CHAR	1		Indicates if this is the last batch of the operation. "J“ last batch "N“ other The value "J" is set if the last output batch is completed when logging off the OP.
INDEX	NUMC	4		Index (counter) of the batches within a production order.
HY_LOSNR	CHAR	20		HYDRA batch number
HY_DLLNR	CHAR	20		HYDRA throughput batch number
Z_MENGE	QUAN	13	3	Not used; by default: 0
Z_MEINH	CHAR	3		Not used; by default: " "
ARBPL	CHAR	8		Producing HYDRA machine
ANZ_TR	NUMC	8		Pallet (package): Number of individual batches for the pallet

Field	T	L	D	Description
LOSHINWEIS	CHAR	20		Entered information on batch
LST01	QUAN	13	3	Activity 1
LST01_EINH	CHAR	3		Unit of activity 1
LST02	QUAN	13	3	Activity 2
LST02_EINH	CHAR	3		Unit of activity 2
LST03	QUAN	13	3	Activity 3
LST03_EINH	CHAR	3		Unit of activity 3
LST04	QUAN	13	3	Activity 4
LST04_EINH	CHAR	3		Unit of activity 4
LST05	QUAN	13	3	Activity 5
LST05_EINH	CHAR	3		Unit of activity 5
LST06	QUAN	13	3	Activity 6
LST06_EINH	CHAR	3		Unit of activity 6
VVDAT	DATE	8		Availability date (MPL-MMO) If MPL-MMO is not used, you should set the value to the current point in time.
VVZEI	TIME	6		Availability time (MPL-MMO) If MPL-MMO is not used, you should set the value to the current point in time.
WDAT	DATE	8		Warning date (MPL-MMO) If MPL-MMO is not used, you should set the value to 31.12.9999.
WZEI	TIME	6		Warning time (MPL-MMO) If MPL-MMO is not used, you should set the value to 23:59:59.
VFDAT	DATE	8		Expiry date (MPL-MMO) If MPL-MMO is not used, you should set the value to 31.12.9999.
VFZEI	TIME	6		Expiry time (MPL-MMO) If MPL-MMO is not used, you should set the value to 23:59:59.
KLASSE	CHAR	1		Batch class "G" Yield "A" Scrap/ waste As of MPL 8.2 the following, additional indicators are available: "O“ Open quantity / problem quantity "N“ Rework
STATUS	CHAR	1		Batch status to be set F Free/available S Blocked
MATST	CHAR	1		Material status
QST	CHAR	1		Quality status
QSTMANU	CHAR	1		Manual quality status
TST	CHAR	1		Transport status

Field	T	L	D	Description
PERSNO	NUM	8		Personnel number
TPE	CHAR	10		Transport unit
CNR_ALT1	CHAR	20		Alternative batch number 1
CNR_ALT2	CHAR	20		Alternative batch number 2
CNR_ALT3	CHAR	20		Alternative batch number 3
CNR_ALT4	CHAR	20		Alternative batch number 4
CNR_ALT5	CHAR	40		Alternative batch number 5
EXTCNR	CHAR	20		External batch number (e.g. batch number)
MCNR	CHAR	20		Merged batch number
ATTR1	NUM	8		Direct batch attribute 1
ATTR2	NUM	8		Direct batch attribute 2
ATTR3	NUM	8		Direct batch attribute 3
ATTR4	DEC	13	3	Direct batch attribute 4
ATTR5	DEC	13	3	Direct batch attribute 5
ATTR6	DEC	13	3	Direct batch attribute 6
ATTR7	CHAR	4		Direct batch attribute 7
ATTR8	CHAR	10		Direct batch attribute 8
ATTR9	CHAR	10		Direct batch attribute 9
ATTR10	CHAR	20		Direct batch attribute 10
CHARGE_LONG	CHAR	20		ERP batch number (long) Available from MPL 8.2 on - please also see the following information on field CHARGE_LONG
MSL_VFDATE	DATE	8		MSL expiry date This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc“ is executed.
MSL_VFTIME	TIME	6		MSL expiry time This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc“ is executed.
MSL_PERIOD	NUMC	8		MSL term This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc“ is executed.

Please note: Batch-related fields are only populated if the movements refer to batches.

Information on the fields CHARGE / CHARGE_LONG

The field CHARGE_LONG is only available as of MPL 8.2.

If MPL 8.2 is used, the fields CHARGE and CHARGE_LONG are populated as follows:

CHARGE includes the ERP batch number with the characters 1-10.

CHARGE_LONG includes the ERP batch number with the characters 1-20.

You should use the value of the CHARGE_LONG field for new installations.

Alphanumeric attributes (Z2CNRATT_C000X000)

The following segment transfers the (first 20) alphanumeric batch attributes.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_101	CHAR	40		Alphanumeric batch attribute
ATTRIB_102	CHAR	40		...
ATTRIB_103	CHAR	40		...
ATTRIB_104	CHAR	40		...
ATTRIB_105	CHAR	40		...
ATTRIB_106	CHAR	40		...
ATTRIB_107	CHAR	40		...
ATTRIB_108	CHAR	40		...
ATTRIB_109	CHAR	40		...
ATTRIB_110	CHAR	40		...
ATTRIB_111	CHAR	40		...
ATTRIB_112	CHAR	40		...
ATTRIB_113	CHAR	40		...
ATTRIB_114	CHAR	40		...
ATTRIB_115	CHAR	40		...
ATTRIB_116	CHAR	40		...
ATTRIB_117	CHAR	40		...
ATTRIB_118	CHAR	40		...
ATTRIB_119	CHAR	40		...
ATTRIB_120	CHAR	40		...

Alphanumeric attributes (Z2CNRATT_C001X000)

The following segment transfers the (first 20) alphanumeric batch attributes.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_121	CHAR	40		Alphanumeric batch attribute
ATTRIB_122	CHAR	40		...
ATTRIB_123	CHAR	40		...
ATTRIB_124	CHAR	40		...
ATTRIB_125	CHAR	40		...
ATTRIB_126	CHAR	40		...
ATTRIB_127	CHAR	40		...
ATTRIB_128	CHAR	40		...
ATTRIB_129	CHAR	40		...
ATTRIB_130	CHAR	40		...
ATTRIB_131	CHAR	40		...
ATTRIB_132	CHAR	40		...
ATTRIB_133	CHAR	40		...
ATTRIB_134	CHAR	40		...
ATTRIB_135	CHAR	40		...
ATTRIB_136	CHAR	40		...
ATTRIB_137	CHAR	40		...
ATTRIB_138	CHAR	40		...
ATTRIB_139	CHAR	40		...
ATTRIB_140	CHAR	40		...

Numeric attributes (Z2CNRATT_N000X000) - OBSOLETE

The following segment transfers the numeric batch attributes.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_201	NUMC	8		Integer, numeric batch attribute
ATTRIB_202	NUMC	8		...
ATTRIB_203	NUMC	8		...
ATTRIB_204	NUMC	8		...
ATTRIB_205	NUMC	8		...
ATTRIB_206	NUMC	8		...
ATTRIB_207	NUMC	8		...
ATTRIB_208	NUMC	8		...
ATTRIB_209	NUMC	8		...
ATTRIB_210	NUMC	8		...
ATTRIB_211	NUMC	8		...
ATTRIB_212	NUMC	8		...
ATTRIB_213	NUMC	8		...
ATTRIB_214	NUMC	8		...
ATTRIB_215	NUMC	8		...
ATTRIB_216	NUMC	8		...
ATTRIB_217	NUMC	8		...
ATTRIB_218	NUMC	8		...
ATTRIB_219	NUMC	8		...
ATTRIB_220	NUMC	8		...
ATTRIB_301	DEC	10	3	Decimal, numeric batch attribute
ATTRIB_302	DEC	10	3	...
ATTRIB_303	DEC	10	3	...
ATTRIB_304	DEC	10	3	...
ATTRIB_305	DEC	10	3	...
ATTRIB_306	DEC	10	3	...
ATTRIB_307	DEC	10	3	...
ATTRIB_308	DEC	10	3	...
ATTRIB_309	DEC	10	3	...
ATTRIB_310	DEC	10	3	...
ATTRIB_311	DEC	10	3	...
ATTRIB_312	DEC	10	3	...
ATTRIB_313	DEC	10	3	...
ATTRIB_314	DEC	10	3	...
ATTRIB_315	DEC	10	3	...
ATTRIB_316	DEC	10	3	...
ATTRIB_317	DEC	10	3	...
ATTRIB_318	DEC	10	3	...
ATTRIB_319	DEC	10	3	...
ATTRIB_320	DEC	10	3	...

Numeric attributes (Z2CNRATT_N001X000)

As of version 1.72946 (2015/SP7) of the script *mle_rckmestyp_zwei_out.hsc,* you can use the segment Z2CNRATT_N001X000 to upload/confirm the numeric attributes of a batch.

Depending on the product version in use, you might have to activate the segment manually:

MPL/TRT 8.1	**MPL/TRT 8.2**
You have to enable uploads via the segment Z2CNRATT_N001X000 manually.	For new installations after SP7/2015 the segment is used by default. You have to enable the segment manually for installations prior to that date.

You have to enable the transfer of the segment manually in the HYDRA INI configuration.

Data is read from the HYDRA table ***los_bestand***

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_201	NUMC	8		Integer, numeric batch attribute
ATTRIB_202	NUMC	8		...
ATTRIB_203	NUMC	8		...
ATTRIB_204	NUMC	8		...
ATTRIB_205	NUMC	8		...
ATTRIB_206	NUMC	8		...
ATTRIB_207	NUMC	8		...
ATTRIB_208	NUMC	8		...
ATTRIB_209	NUMC	8		...
ATTRIB_210	NUMC	8		...
ATTRIB_211	NUMC	8		...
ATTRIB_212	NUMC	8		...
ATTRIB_213	NUMC	8		...
ATTRIB_214	NUMC	8		...
ATTRIB_215	NUMC	8		...
ATTRIB_216	NUMC	8		...
ATTRIB_217	NUMC	8		...
ATTRIB_218	NUMC	8		...
ATTRIB_219	NUMC	8		...
ATTRIB_220	NUMC	8		...

Field	T	L	D	Description
ATTRIB_301	DEC	13	3	Decimal, numeric batch attribute
ATTRIB_302	DEC	13	3	...
ATTRIB_303	DEC	13	3	...
ATTRIB_304	DEC	13	3	...
ATTRIB_305	DEC	13	3	...
ATTRIB_306	DEC	13	3	...
ATTRIB_307	DEC	13	3	...
ATTRIB_308	DEC	13	3	...
ATTRIB_309	DEC	13	3	...
ATTRIB_310	DEC	13	3	...
ATTRIB_311	DEC	13	3	...
ATTRIB_312	DEC	13	3	...
ATTRIB_313	DEC	13	3	...
ATTRIB_314	DEC	10	3	...
ATTRIB_315	DEC	10	3	...
ATTRIB_316	DEC	10	3	...
ATTRIB_317	DEC	10	3	...
ATTRIB_318	DEC	10	3	...
ATTRIB_319	DEC	10	3	...
ATTRIB_320	DEC	10	3	...

Tree of generation (Z2CNRBAUM000X000)

The following data record is part of the tree of generation of a batch. These are "OPTIONAL“ segments, i.e. they are only included in the IDoc if they actually exist.

Data is read from the HYDRA table ***LOS_ZUORDNUNGEN***.

Field	T	L	D	Description
HY_ALOSNR lz.al_nr	CHAR	20		HYDRA batch number (output batch)
A_CHARGE lz.al_charge	CHAR	10		ERP batch number of the output batch ... populated if the batch is an ERP batch.
A_MATNR lz.al_matnr	CHAR	40		Material number of the output batch Known in the ERP system for inventory-managed materials.
HY_ELOSNR	CHAR	20		HYDRA batch number (input batch)
E_CHARGE	CHAR	10		ERP batch number of the input batch ... populated if the batch is an ERP batch.
E_MATNR	CHAR	40		Material number of the input batch Known in the ERP system for inventory-managed materials.
E_POS	CHAR	10		BOM item of the input batch
ANR	CHAR	40		HYDRA production order producing the output batch.

Field	T	L	D	Description
ARBPL	CHAR	8		Machine where the output batch was generated.
DATUM	DATE	8		Creation date of the output batch
ZEIT	TIME	6		Creation time of the output batch

Note

You should manage this information in a user-specific table "ZCNRBAUM“ in the ERP system. The structure of this table should be identical to that of the segment.

Sub-batches (Z2TOLO000X000)

For pallets: The described sub-segment transfers the included sub-batches that are assigned to the pallets. These are "OPTIONAL“ segments, i.e. they are only included in the IDoc if they actually exist.

Data is read from the HYDRA table ***ZTOLO***.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number of individual batch
MATNR	CHAR	40		HYDRA material number
MATTYP	CHAR	10		HYDRA material type
MATTXT	CHAR	40		HYDRA material name
MENGE	QUAN	13	3	Quantity of individual batch
MENGE_EINH	CHAR	3		Quantity unit
LST01	QUAN	13	3	Activity 1
LST01_EINH	CHAR	3		Unit of activity 1
LST02	QUAN	13	3	Activity 2
LST02_EINH	CHAR	3		Unit of activity 2
LST03	QUAN	13	3	Activity 3
LST03_EINH	CHAR	3		Unit of activity 3
LST04	QUAN	13	3	Activity 4
LST04_EINH	CHAR	3		Unit of activity 4
LST05	QUAN	13	3	Activity 5
LST05_EINH	CHAR	3		Unit of activity 5
LST06	QUAN	13	3	Activity 6
LST06_EINH	CHAR	3		Unit of activity 6

User fields (Z2CNR_USRFLD000X000)

You can transfer/upload the user fields of a batch as of version 1.8 of the script *mle_rckmestyp_zwei_out.hsc.* Data is transferred in the segment Z2CNR_USRFLD000X000.

You have to enable the transfer of the segment manually in the HYDRA INI configuration.

Data is read from the HYDRA table ***los_bestand***

Field	V	T	L	D	Description
HY_LOSNR	S	CHAR	20		HYDRA batch number
FILLER	S	CHAR	20		Placeholder – internal usage
USRFLD	S	CHAR	8		User field key
FU:1	K	DATE	8		User field 1
FU:2	K	DATE	8		User field 2
FU:3	K	DATE	8		User field 3
FU:4	K	DATE	8		User field 4
FU:5	K	DATE	8		User field 5
FU:6	K	DATE	8		User field 6
FU:7	K	NUM	8		User field 7
FU:8	K	NUM	8		User field 8
FU:9	K	NUM	8		User field 9
FU:10	K	NUM	8		User field 10
FU:11	K	NUM	8		User field 11
FU:12	K	NUM	8		User field 12
FU:13	K	NUM	8		User field 13
FU:14	K	NUM	8		User field 14
FU:15	K	NUM	8		User field 15
FU:16	K	NUM	8		User field 16
FU:17	K	NUM	8		User field 17
FU:18	K	NUM	8		User field 18
FU:19	K	NUM	8		User field 19
FU:20	K	NUM	8		User field 20
FU:21	K	NUM	8		User field 21
FU:22	K	NUM	8		User field 22
FU:23	K	DEC	13	3	User field 23
FU:24	K	DEC	13	3	User field 24
FU:25	K	DEC	13	3	User field 25
FU:26	K	DEC	13	3	User field 26
FU:27	K	DEC	13	3	User field 27
FU:28	K	DEC	13	3	User field 28
FU:29	K	CHAR	1		User field 29
FU:30	K	CHAR	1		User field 30
FU:31	K	CHAR	1		User field 31
FU:32	K	CHAR	1		User field 32
FU:33	K	CHAR	1		User field 33
FU:34	K	CHAR	1		User field 34

Field	V	T	L	D	Description
FU:35	K	CHAR	1		User field 35
FU:36	K	CHAR	1		User field 36
FU:37	K	CHAR	1		User field 37
FU:38	K	CHAR	1		User field 38
FU:39	K	CHAR	1		User field 39
FU:40	K	CHAR	1		User field 40
FU:41	K	CHAR	1		User field 41
FU:42	K	CHAR	1		User field 42
FU:43	K	CHAR	1		User field 43
FU:44	K	CHAR	1		User field 44
FU:45	K	CHAR	10		User field 45
FU:46	K	CHAR	10		User field 46
FU:47	K	CHAR	10		User field 47
FU:48	K	CHAR	10		User field 48
FU:49	K	CHAR	10		User field 49
FU:50	K	CHAR	10		User field 50
FU:51	K	CHAR	20		User field 51
FU:52	K	CHAR	20		User field 52
FU:53	K	CHAR	20		User field 53
FU:54	K	CHAR	20		User field 54
FU:55	K	CHAR	20		User field 55
FU:56	K	CHAR	20		User field 56
FU:57	K	CHAR	20		User field 57
FU:58	K	CHAR	20		User field 58
FU:59	K	CHAR	20		User field 59
FU:60	K	CHAR	20		User field 60
FU:61	K	CHAR	20		User field 61
FU:62	K	CHAR	20		User field 62
FU:63	K	CHAR	20		User field 63
FU:64	K	CHAR	20		User field 64
FU:65	K	CHAR	40		User field 65
FU:66	K	CHAR	40		User field 66

7 Usage Decision MES --> ERP

Summary

The collection in HYDRA transforms the production orders into output batches that will be transferred as goods receipt for the material, which is inventory-managed in the PPS system.

For each lot (= batch) it is possible in HYDRA to take a usage decision and to transmit it to the PPS-system.

To do so, the option "Retain until usage decision“ will be used within the material type of the operation. After that a usage decision was taken for a batch, the goods receipt (ZWEI) will be confirmed/uploaded and if necessary also the usage decision with the corresponding code.

The data is read from the HYDRA table ***EVENT_LOS***.

For the IDOC this leads to the following specification:

Message type:	**ZCNRVEW**
File name (with file-transfer):	**Z2CNRVEW000X000**
File extension (with file transfer):	Subject to configuration (by default .dat“)
IDOC type:	**ZCNRVEW02**
Segments:	**Z2CNRVEW000X000** (usage decision)

Usage decision (Z2CNRVEW000X000)

This structure is used to transfer the usage decision taken in HYDRA to a batch (a HYDRA batch).

Field	T	L		Description
WERK	CHAR	4		Company/ plant; stored fix to HYDRA
ANR	CHAR	40		Production order that has produced
SAP_AUNR	12			SAP order number
SAP_AFOLG	6			SAP sequence number
SAP_VORNR	4			SAP operation number
SAP_UVGNR	4			SAP sub-operation number
CHARGE	CHAR	10		PPS batch number

Field	T	L		Description
HY_LOSNR	CHAR	20		HYDRA batch number
DATUM	DATE	8		Date of the usage decision (format yyyymmdd)
ZEIT	TIME	6		Time of the usage decision (format hhmmss)
VCODE	CHAR	4		Usage decision - code "RFB“ Free stock "RSB“ Locked stock "AUS“ Scrap
VCODEGRP	CHAR	8		Usage decision – code group fix: " “ (blanks)

8 Settings Relevant to the Application in HYDRA

Maintenance of the distribution model - HYDRA inbound processing

Use the HYDRA distribution model to maintain entries for HYDRA inbound processing:

Parameter name	**Value**
To process material staging/material supply	
Message type	ZMBEW
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=<MLE variant to be used
Description	Material staging ERP → HYDRA
Log. target system	Created logical system
Storage duration	10

Please restart HYDRA after editing entries.

Maintenance of the distribution model - HYDRA outbound processing

Use the HYDRA distribution model to maintain entries for HYDRA outbound processing:

Parameter name	**Value**
To upload material withdrawals (consumptions):	
Message type	ZWAU
Description	Upload material withdrawals
IDoc type	ZWAU

Parameter name	Value
Storage duration	10
Log. target system	Created logical system
Segment name 1	Z2WAU000X000
To upload incoming materials:	
Message type	ZWEI
Description	Upload incoming materials
IDoc type	ZWEI02
Storage duration	10
Log. target system	Created logical system
Segment name 1	Z2WEI000X000
To upload the usage decision	
Message type	ZCNRVEW
Description	Upload of the usage decision
IDoc type	ZCNRVEW02
Storage duration	10
Log. target system	Created logical system
Segment name 1	Z2CNRVEW000X000

Scheduler maintenance

The following entries must be made for confirmations/uploads of goods movements in the Scheduler:

Parameter name	Value
To upload incoming materials:	

Parameter name	Value
Product key	MPL-BP
License key	MPL-BP
Command (Windows):	sh.exe ./myerprck.scr /MESTYP=ZWEI
Command (Unix):	./myerprck.scr /MESTYP=ZWEI
Comment:	Goods receipt HYDRA → ERP
Interval	5
To upload material consumptions:	
Product key	MPL-BP
License key	MPL-BP
Command (Windows):	sh.exe ./myerprck.scr /MESTYP=ZWAU
Command (Unix):	./myerprck.scr /MESTYP=ZWAU
Comment:	Goods issue HYDRA → ERP
Interval	5
To upload the usage decision:	
Product key	MPL-BP
License key	MPL-BP
Command (Windows):	sh.exe ./myerprck.scr /MESTYP=ZCNRVEW
Command (Unix):	./myerprck.scr /MESTYP=ZCNRVEW
Comment:	Usage decision HYDRA → ERP
Interval	5
To upload incoming material:	
Product key	MPL-BP

Parameter name	Value
License key	MPL-BP
Command (Windows):	sh.exe ./hysapupl.scr /UPLSEGNAM=Z2WEI000X000 /SINGLE_IDOC /SUBLEVEL=2
Command (Unix):	./hysapupl.scr /UPLSEGNAM=Z2WEI000X000 /SINGLE_IDOC /SUBLEVEL=2
Comment:	Upload of incoming goods HYDRA → ERP
Interval	5
To upload outgoing material:	
Product key	MPL-BP
License key	MPL-BP
Command (Windows):	sh.exe ./hysapupl.scr /UPLSEGNAM=Z2WAU000X000
Command (Unix):	./hysapupl.scr /UPLSEGNAM=Z2WAU000X000
Comment:	Upload goods issues HYDRA → ERP
Interval	5
To upload the usage decision	
Product key	MPL-BP
License key	MPL-BP
Command (Windows):	sh.exe ./hysapupl.scr /UPLSEGNAM=Z2CNRVEW000X000
Command (Unix):	./hysapupl.scr /UPLSEGNAM=Z2CNRVEW000X000
Comment:	Upload goods issues HYDRA → ERP

Parameter name	Value
Interval	5

Activation in material type

Set the indicator *Goods movements > Transfer to interface* for the material types, for which a transfer of the material movements is necessary.

In case the HYDRA material type is not available as application, the indicator can also be set directly via the database:

```
update hz_typen set we_ext_kz = ,J‘ where hz_typ = ‘<Material type for which the indicator is to be set>’;
```

INI-Configuration for segment Z2CNR_USRFLD000X000

Provided with version 1.8 of the script *mle_rckmestyp_zwei_out.hsc* there is the option to transfer/upload the user fields of a batch as well. The transfer/upload has to be activated explicitly in the HYDRA INI configuration.

Parameter name	Value
INI name	MCL
Section	ZWEI
Key	USERFIELDS
Value	J / Y Transfer of the user field segment
Active	Ja (yes)
Comment	Transfer of the segment Z2CNR_USRFLD000X000

INI configuration for the segment Z2CNRATT_N001X000

Provided with version 1.72946 of the script *mle_rckmestyp_zwei_out.hsc* there is the option to use Z2CNRATT_N001X000 to upload numeric batch attributes. Depending on the product version in use, the segment has to be activated manually in HYDRA INI configuration:

MPL/TRT 8.1	**MPL/TRT 8.2**
Uploads via segment Z2CNRATT_N001X000 must be enabled manually.	For new installations after SP7/2015 the segment is used by default. For installations prior to that date the segment has to be enabled manually.

The transfer/upload has to be activated explicitly in the HYDRA INI configuration.

Parameter name	**Value**
INI name	MCL
Section	ZWEI
Key	CNRATTR_DEC_13_3
Value	J / Y Transfer of the segment Z2CNRATT_N001X000
Active	Ja (yes)
Comment	Transfer of the segment Z2CNRATT_N001X000

9 Test Files

Overview

Attached to this documentation, you will find test files for the interface EIS-MCL. The attachment is only available, if the documentation is in PDF format.

The documentation Open PDF attachments describes how to call the attached test files.

The following test files are attached to the PDF document:

File	Type	Comment
Z2WAU000X000.dat	Outbound processing	Sample file for the material consumption MES --> ERP in HYDRA standard format
Z2WEI000X000.dat	Outbound processing	Sample file for the goods receipt MES --> ERP in HYDRA standard format
ZMBEW.dat	Inbound processing	Sample file for the material staging ERP --> MES in HYDRA standard format